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AHEAD Online

Object Oriented Programming Using Java

Objectives

- Inheritance.

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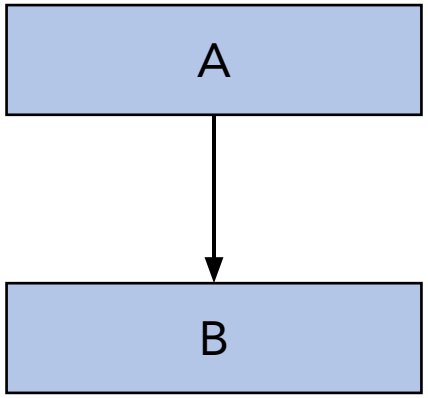
Inheritance

- Inheritance is one of the cornerstones of object-oriented programming.
- It allows the creation of hierarchical classifications.
- Using inheritance, we can create a general class that defines traits common to a set of related items.
- Inheritance supports the concept of "reusability".

Inheritance

- A class that is inherited is called a *superclass*.
- The class that does the inheriting is called a *subclass.*
- a subclass is a specialized version of a superclass.
- It inherits all of the instance variables and methods defined by the superclass and adds its own, unique elements.



Inheritance

The general form of a class declaration that inherits a superclass

class subclass-name extends superclass-name {

//Body of class

}

Example:

Class B extends A{}

B – Subclass

A - Superclass

Example

```
// Create a superclass.
class A
{
int i, j;
void showij()
{
    System.out.println("i and j: " + i + " " + j);
}
}
```

```
class B extends A
{
int k;
void showk()
{
   System.out.println("k: " + k);
}
void sum()
{
    System.out.println("i+j+k: " + (i+j+k));
}
}
```

Properties of subclass

- Subclass includes all of the members of its superclass, it cannot access members of the superclass declared as private.
- Subclass can access private members of super class through its public methods.
- Constructors are not inherited by subclasses, but the constructor of the superclass can be invoked from the subclass.

Properties of subclass

- A subclass can have only one superclass(Java does not support multiple inheritance).
- We can declare new methods in the subclass that are not in the superclass.
- We can declare methods in the subclass that as the same signature as the one in the superclass.

Types of Inheritance

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Hybrid Inheritance.

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Takeaways

- Inheritance.
- Superclass and subclass.
- Types of Inheritance.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*